

AI Awareness Workshop

Comprehensive Event Report & Technical Proceedings

DATE 25 August 2026	VENUE Seminar Hall
DEPARTMENT Computer Science	TARGET AUDIENCE Undergraduate & Graduate Students, Faculty

1. Executive Summary

The Department of Computer Science successfully hosted the **AI Awareness Workshop** on 25 August 2026 at the institutional Seminar Hall. The event was designed to demystify artificial intelligence, machine learning, and modern neural network paradigms for students and academic staff. Through a structured blend of keynote presentations, live code walkthroughs, ethical impact discussions, and interactive Q&A sessions, attendees gained deep insights into current trends and future trajectories in intelligent systems.

Key Milestone: Over 185 active participants engaged with hands-on demonstrations showcasing transformer architectures, generative AI toolchains, and real-world deployment frameworks.

2. Detailed Event Schedule & Agenda

The workshop span a full day, categorized into morning theoretical foundations and afternoon practical implementation modules.

Time Slot	Session Title	Key Focus Areas & Deliverables
09:30 AM – 10:00 AM	Inauguration & Keynote Note	Introduction by Head of Department; overview of AI evolution.
10:00 AM – 11:30 AM	Foundations of Neural Networks	Perceptrons, backpropagation, activation functions, and gradient descent ($J(\theta)$).
11:30 AM – 11:45 AM	Networking & Tea Break	Interactive networking session among student delegates and faculty.
11:45 AM – 01:00 PM	Modern GenAI & Large Models	Transformer models, attention mechanisms, tokenization, and embeddings.
	Hands-On Python Coding Lab	

Time Slot	Session Title	Key Focus Areas & Deliverables
02:00 PM – 03:30 PM	Ethics, Safety & Future Outlook	Building lightweight classifiers using PyTorch and Scikit-Learn.
03:30 PM – 04:30 PM		Bias mitigation, data privacy regulations, copyright, and alignment.
04:30 PM – 05:00 PM	Valediction & Certificate Distribution	Feedback collection, quiz results announcement, and closing remarks.

3. Core Technical Modules Covered

3.1 Mathematical Foundations of Optimization

A rigorous review of optimization algorithms utilized in training deep learning models was presented. Special emphasis was placed on loss landscapes and convergence rates. The objective function minimization is formally represented as:

$$\theta^* = \arg \min_{\theta} \ell(f(x_i; \theta), y_i) + \lambda R(\theta)$$

Where ℓ represents the empirical loss, f denotes the model prediction, and $R(\theta)$ ensures regularization to prevent overfitting across high-dimensional feature spaces.

3.2 Natural Language Processing & Transformers

Participants explored self-attention mathematics, fundamental to modern language models. The scaled dot-product attention score is computed via:

$$Attention(Q, K, V) = softmax(\{QK^T / \sqrt{d_k}\}) V$$

The session highlighted how parallel matrix multiplications drastically accelerate training epochs compared to traditional recurrent neural networks (RNNs).

4. Participant Demographics & Engagement Statistics

The workshop witnessed enthusiastic participation across diverse academic levels within the Department of Computer Science.

Participant Category	Registered Count	Attendance Rate	Feedback Rating (Out of 5.0)
Undergraduate Students (B.Sc. / B.Tech)	110	94%	4.8

Participant Category	Registered Count	Attendance Rate	Feedback Rating (Out of 5.0)
Postgraduate Students (M.Sc. / M.Tech)	55	90%	4.9
Research Scholars & Ph.D. Fellows	20	95%	4.7
Faculty & External Delegates	15	86%	4.9

5. Key Takeaways & Recommendations

- **Curriculum Integration:** Faculty members agreed to introduce elective modules on Large Language Model internals and prompt engineering principles for the upcoming semester.
- **Infrastructure Upgrade:** Recommendations were put forward to expand GPU cluster access in the departmental computing lab to support student research projects.
- **Student Coding Club:** A bi-weekly AI coding circle was established to foster peer-to-peer collaboration on open-source machine learning repositories.

6. Conclusion

The AI Awareness Workshop successfully accomplished its core objective of raising technical awareness and fostering responsible innovation mindsets among students. The active engagement observed throughout the sessions in the Seminar Hall reaffirms the Department of Computer Science's commitment to staying at the forefront of technological education.